

MIS 443 – FINAL EXAMINATION

Finance Analysis with PostgreSQL

Course	MIS 443	Business Data Management
Duration	90 minutes	Maximum mark: 100

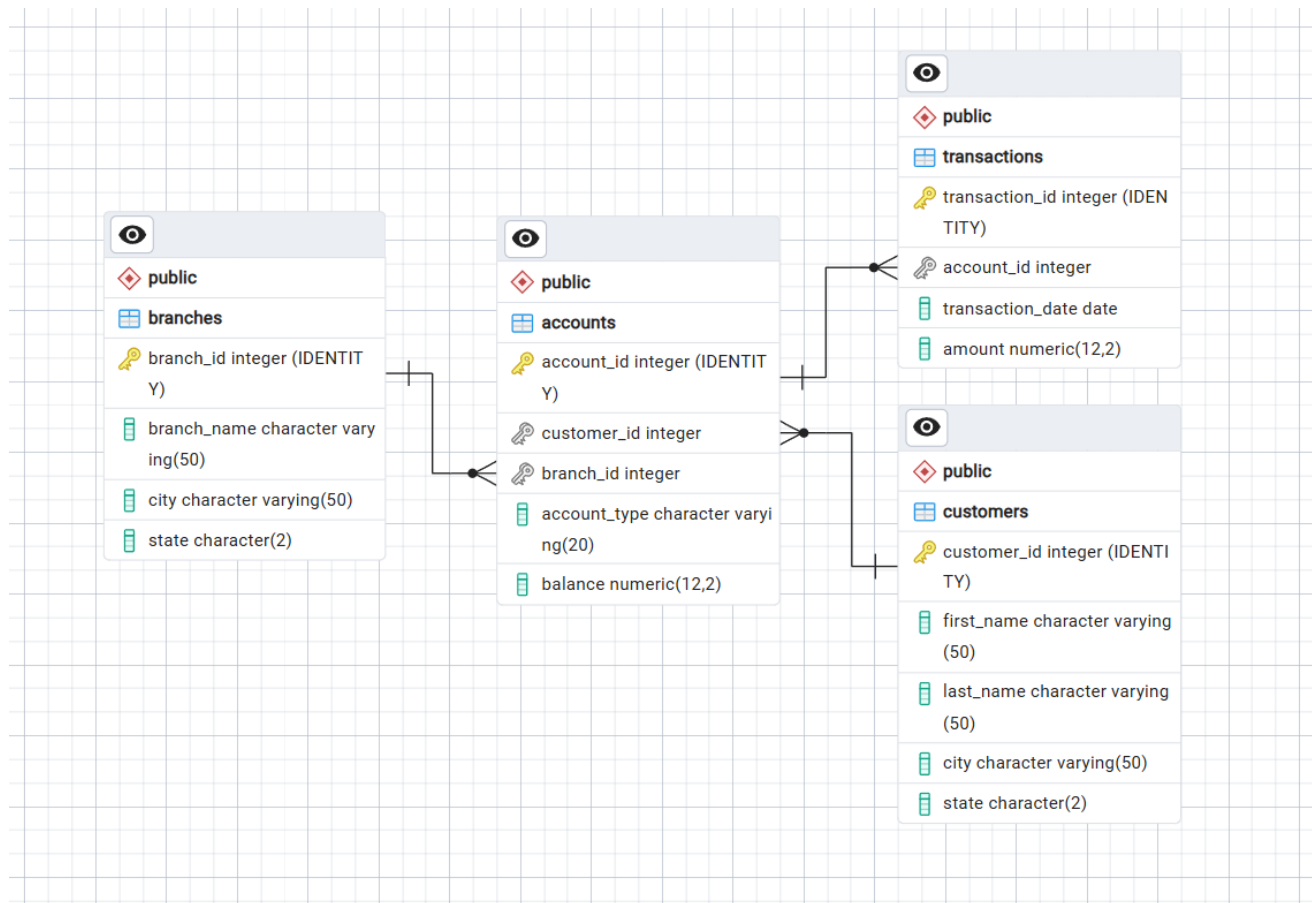
Dataset Introduction

The dataset represents a retail bank that manages customers, branches, bank accounts, and financial transactions. Managers use the data to analyse customer portfolios, account balances, branch performance, and transaction activity.

- Customers: stores customer names and locations.
- Branches: stores branch names and locations.
- Accounts: stores account ownership, branch assignment, account type, and current balance.
- Transactions: stores the date and amount of each account transaction.

Each account belongs to one customer and one branch, and each account may have multiple transactions. Positive account balances represent funds held by customers; negative Credit Card balances represent amounts owed. Positive transaction amounts are credits, while negative amounts are debits.

Entity–Relationship Diagram



Examination Questions

Question 1 – Database Setup (10 marks)

Create a PostgreSQL database using your full name in lowercase, without spaces or Vietnamese diacritics. Connect to the database and execute `MIS443_Finance_PostgreSQL.sql`. Confirm that the customers, branches, accounts, and transactions tables are available in the public schema.

Question 2 – Customer and Account Overview (10 marks)

- (a) The Customer Service Manager needs a contact list of customers living in New York. Display `customer_id`, the customer's full name as `full_name`, and city. Sort by `customer_id`. (5 marks)

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including a server named 'PostgreSQL 17' with a database 'vanvuquynhnhu'. The main pane shows a SQL query window with the following text:

```

/* Question 2 - Customer and Account Overview (10 marks)
(a) The Customer Service Manager needs a contact list of customers living in New York.
Display customer_id, the customer's full name as full_name, and city. Sort by customer_id. (5 marks)
*/
select c.customer_id, c.first_name, c.last_name, c.city
from customers c
where c.city = 'New York';

```

The 'Data Output' tab shows the results of the query:

customer_id [PK] integer	first_name character varying (50)	last_name character varying (50)	city character varying (50)
1	John	Doe	New York
2	Jane	Doe	New York

The status bar at the bottom indicates 'Total rows: 2' and 'Query complete 00:00:00.275'.

(b) Management needs to confirm the size of the account portfolio. Calculate the total number of accounts and name the result total_accounts. (5 marks)

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including a server named 'PostgreSQL 17' with a database 'vanvuquynhnhu'. The main pane shows a SQL query window with the following text:

```

/* Question 2 - Customer and Account Overview (10 marks)
(b) Management needs to confirm the size of the account portfolio.
Calculate the total number of accounts and name the result total_accounts. (5 marks)
*/
select count(a.account_id) as total_accounts
from accounts a;

```

The 'Data Output' tab shows the results of the query:

total_accounts bigint
15

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.078'.

Question 3 – Account Balance Analysis (20 marks)

(a) The Finance Manager wants to monitor funds held in Checking accounts. Calculate their total balance and name the result total_checking_balance. (10 marks)

pgAdmin 4

Query: Query History

```

/* Question 3 - Account Balance Analysis (20 marks)
(a) The Finance Manager wants to monitor funds held in Checking accounts.
Calculate their total balance and name the result total_checking_balance. (10 marks)
*/

select sum(a.balance) as total_checking_balance
from accounts a
where a.account_type = 'Checking';

```

Data Output Messages Notifications

	total_checking_balance numeric
1	31000.00

Total rows: 1 Query complete 00:00:00.096

(b) For each customer living in Los Angeles, display customer_id, full_name, and total_balance across all account types. Sort by total_balance from highest to lowest. (10 marks)

pgAdmin 4

Query: Query History

```

/* Question 3 - Account Balance Analysis (20 marks)
For each customer living in Los Angeles, display customer_id, full_name, and total_balance across all account types.
Sort by total_balance from highest to lowest. (10 marks)
*/

select c.customer_id,
       c.first_name || ' ' || c.last_name as full_name,
       sum(a.balance) as total_balance
from customers c
inner join accounts a on a.customer_id = c.customer_id
where c.city = 'Los Angeles'
group by c.customer_id,
         full_name
order by total_balance desc
;

```

Data Output Messages Notifications

	customer_id [PK] integer	full_name text	total_balance numeric
1	5	Michael Lee	69000.00
2	6	Jennifer Wa...	15000.00

Total rows: 2 Query complete 00:00:00.130

Question 4 – Branch and Customer Portfolio Analysis (20 marks)

(a) Identify the branch with the highest average account balance. Display branch_id, branch_name, city, and average_balance. Round to two decimal places and include ties. (10 marks)

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'vanvuquynhhu'. The main window shows a SQL query for Question 4(a):

```

/*
Question 4 – Branch and Customer Portfolio Analysis (20 marks)
(a) Identify the branch with the highest average account balance.
Display branch_id, branch_name, city, and average_balance. Round to two decimal places and include ties. (10 marks)
*/
select b.branch_id,
       b.branch_name,
       b.city,
       round(avg(a.balance),2) as average_account_balance
from branches b
inner join accounts a on a.branch_id = b.branch_id
group by b.branch_id, b.branch_name, b.city
order by average_account_balance desc ;

```

The query results are displayed in a table with 5 rows:

branch_id	branch_name	city	average_account_balance
14	North Beach	San Francisco	30000.00
8	South Bay	San Francisco	12500.00
2	Downtown	San Francisco	7100.00
1	Main	New York	3833.33
5	Uptown	New York	1833.33

The status bar at the bottom indicates 'Total rows: 5' and 'Query complete 00:00:00.112'.

(b) Identify the customer who owns the single account with the highest current balance. Display customer_id, full_name, account_id, account_type, and balance. Include ties. (10 marks)

The screenshot shows the pgAdmin 4 interface. The main window shows a SQL query for Question 4(b):

```

/*
Question 4 – Branch and Customer Portfolio Analysis (20 marks)
(b) Identify the customer who owns the single account with the highest current balance.
Display customer_id, full_name, account_id, account_type, and balance. Include ties. (10 marks)
*/
select c.customer_id,
       c.first_name || ' ' || c.last_name as full_name,
       a.account_id,
       a.account_type,
       a.balance
from customers c
inner join accounts a on a.customer_id = c.customer_id
group by c.customer_id, full_name, a.account_id, a.account_type, a.balance
order by a.balance desc
limit 1;

```

The query results are displayed in a table with 1 row:

customer_id	full_name	account_id	account_type	balance
5	Michael Lee	10	Savings	30000.00

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.079'.

Question 5 – Customer Value and Activity (20 marks)

(a) Identify the most active customers based on the total number of transactions across all their accounts. Display customer_id, full_name, and total_transactions. Include ties. (10 marks)

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'vanvuquynhnhu' on 'PostgreSQL 17'. The main pane shows a SQL query for 'Question 5 - Customer Value and Activity (20 marks)'. The query is:


```

  /*
  Question 5 - Customer Value and Activity (20 marks)
  (a) Identify the most active customers based on the total number of transactions across all their accounts.
  Display customer_id, full_name, and total_transactions. Include ties. (10 marks)
  */
  select c.customer_id,
         c.first_name || ' ' || c.last_name as full_name,
         count(t.transaction_id) as total_transactions
  from customers c
  inner join accounts a on a.customer_id = c.customer_id
  inner join transactions t on a.account_id = t.account_id
  group by c.customer_id, full_name
  order by total_transactions desc;
  
```

 The 'Data Output' pane shows the results of the query:

customer_id [PK] integer	full_name text	total_transactions bigint
1	2 Jane Doe	4
2	4 Alice Johns...	4
3	3 Bob Smith	3
4	1 John Doe	3
5	5 Michael Lee	1

 The status bar at the bottom indicates 'Total rows: 5' and 'Query complete 00:00:00.088'.

- (b) Identify the customer with the highest combined balance across Checking and Savings accounts. Exclude Credit Card accounts. Display customer_id, full_name, and total_deposit_balance. Include ties. (10 marks)

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'vanvuquynhnhu' on 'PostgreSQL 17'. The main pane shows a SQL query for 'Question 5 - Customer Value and Activity (20 marks)'. The query is:


```

  /*
  Question 5 - Customer Value and Activity (20 marks)
  b) Identify the customer with the highest combined balance across Checking and Savings accounts.
  Exclude Credit Card accounts. Display customer_id, full_name, and total_deposit_balance. Include ties. (10 marks)
  */
  select c.customer_id,
         c.first_name || ' ' || c.last_name as full_name,
         sum(a.balance) as total_deposit_balance
  from customers c
  inner join accounts a on a.customer_id = c.customer_id
  where a.account_type = 'Checking'
     or a.account_type = 'Savings'
  group by c.customer_id, full_name
  order by total_deposit_balance desc;
  
```

 The 'Data Output' pane shows the results of the query:

customer_id [PK] integer	full_name text	total_deposit_balance numeric
1	5 Michael Lee	60000.00
2	4 Alice Johns...	25000.00
3	3 Bob Smith	22500.00
4	6 Jennifer Wa...	15000.00
5	2 Jane Doe	12500.00
6	1 John Doe	6000.00

 The status bar at the bottom indicates 'Total rows: 6' and 'Query complete 00:00:00.075'.

Question 6 – Advanced Finance Analysis (20 marks)

- (a) Identify the branch with the highest total balance across all account types. Display branch_id, branch_name, and total_balance. Include ties. (10 marks)

The screenshot shows the pgAdmin 4 interface. The SQL query window contains the following code:

```

--
-- Question 6 - Advanced Finance Analysis (20 marks)
-- (a) Identify the branch with the highest total balance across all account types.
-- Display branch_id, branch_name, and total_balance. Include ties. (10 marks)
--
select b.branch_id, b.branch_name, sum(balance) as total_balance
from branches b
inner join accounts a on a.branch_id = b.branch_id
group by b.branch_id, b.branch_name
order by total_balance desc;

```

The Results window shows the following data:

branch_id (PK) integer	branch_name character varying (50)	total_balance numeric
1	North Beach	60000.00
2	Downtown	25000.00
3	South Bay	25000.00
4	Main	15000.00
5	Uptown	5500.00

Total rows: 5

- (b) Rank all customers by total balance across all account types. Equal totals must receive the same rank without gaps. Display customer_id, full_name, total_balance, and balance_rank. Do not use a CTE. (5 marks)

The screenshot shows the pgAdmin 4 interface. The SQL query window contains the following code:

```

--
-- Question 6 - Advanced Finance Analysis (20 marks)
-- (b) Rank all customers by total balance across all account types.
-- Equal totals must receive the same rank without gaps. Display customer_id, full_name, total_balance, and balance_rank.
-- Do not use a CTE. (5 marks)
--
select c.customer_id,
       c.first_name || ' ' || c.last_name as full_name,
       sum(a.balance) as total_balance,
       dense_rank() over (
         order by sum(a.balance) desc
       ) as balance_rank
from customers c
inner join accounts a on a.customer_id = c.customer_id
group by c.customer_id, full_name
order by balance_rank, c.customer_id;

```

The Results window shows the following data:

customer_id (PK) integer	full_name text	total_balance numeric	balance_rank bigint
5	Michael Lee	60000.00	1
4	Alice Johns...	25000.00	2
3	Bob Smith	25000.00	3
6	Jennifer Wa...	15000.00	4
2	Jane Doe	11500.00	5
1	John Doe	5500.00	6

Total rows: 6

- (c) Use a CTE to calculate the total number of transactions for every branch, including branches with no transactions. Return the branch or branches with the highest total. Display branch_id, branch_name, and total_transactions. (5 marks)

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including a database named 'vanvuquynhnhu' with several tables. The main pane shows a SQL query window with the following text:

```

(c) Use a CTE to calculate the total number of transactions for every branch, including branches with no transactions.
Return the branch or branches with the highest total. Display branch_id, branch_name, and total_transactions. (5 marks)
*/

with branch_transaction as (
  select b.branch_id, b.branch_name, count(t.transaction_id) as total_transactions
  from branches b
  inner join accounts a on a.branch_id = b.branch_id
  inner join transactions t on t.account_id = a.account_id
  group by b.branch_id, b.branch_name
)

select branch_id, branch_name, total_transactions
from branch_transaction
order by total_transactions desc

```

Below the query window, the 'Data Output' tab shows the results of the query:

branch_id	branch_name	total_transactions
1	Main	4
2	South Bay	4
3	Uptown	3
4	Downtown	3
5	North Beach	1

At the bottom, a status bar indicates 'Total rows: 5' and 'Query complete 00:00:00.075'. A green message box at the bottom right states: 'Successfully run. Total query runtime: 75 msec. 5 rows affected.'

Mark Distribution

Question	Marks
Question 1	10
Question 2	10
Question 3	20
Question 4	20
Question 5	20
Question 6	20

END OF EXAMINATION